Here is the complete project documentation for your business profit and cost analysis program, structured just like the previous ones!

1. Problem Statement, Objectives, and Assumptions

Problem Statement

Manual financial tracking in manufacturing or small-scale production—combining material, labor, and machine costs against total revenue and net profit—is prone to human calculation errors. Businesses require an automated financial analysis tool to instantly compute production expenses, aggregate revenues, net profits, and evaluate performance against a predefined profit target.

Objectives

- **Automate Cost Aggregation:** To combine material, labor, and machine costs into a single comprehensive production cost.
- **Calculate Revenue & Profit:** To compute total sales revenue based on output quantity and unit price, and derive the net profit.
- **Target Evaluation:** To automatically verify whether the generated profit meets or exceeds the organizational benchmark (Rs.$5{,}000$).

Assumptions

- Financial inputs (costs and selling price) are numeric values.
- Production quantity is provided as a whole number (integer).

- The minimum profit threshold is fixed at Rs.$5{,}000$.

2. Algorithm, Flowchart, and Pseudocode

Algorithm

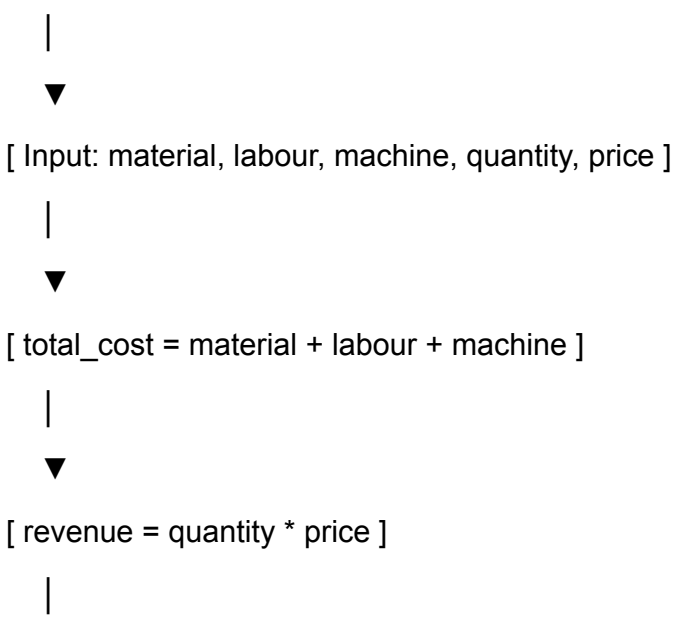
1. **Start** the program.
2. **Input** material cost, labour cost, and machine cost as floating-point numbers.
3. **Input** quantity produced as an integer and unit price as a floating-point number.
4. **Calculate Total Cost:** total_cost = material + labour + machine.
5. **Calculate Revenue:** revenue = quantity * price.
6. **Calculate Profit:** profit = revenue - total_cost.
7. **Display Results:** Print Total Cost, Revenue, and Profit amounts.
8. **Evaluate Profit Target:**

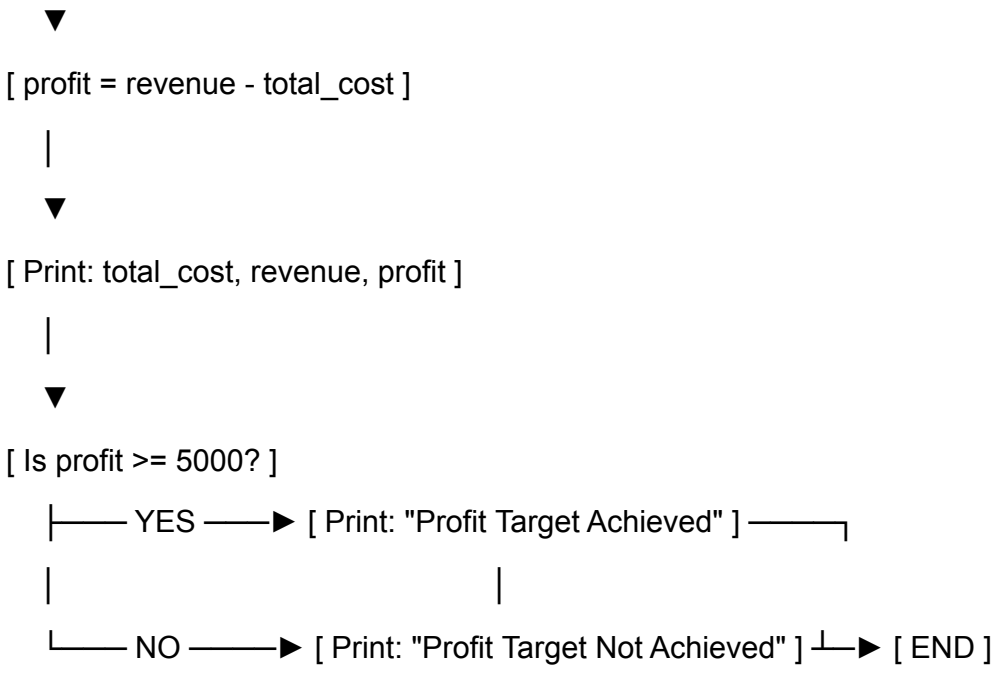
- If profit >= 5000, print "Profit Target Achieved".
- Else, print "Profit Target Not Achieved".

9. **End** the program.

Flowchart Representation

Plaintext





Pseudocode

Plaintext

```
BEGIN
    PRINT "Enter material cost: "
    READ material
    PRINT "Enter labour cost: "
    READ labour
    PRINT "Enter machine cost: "
    READ machine
```

```
    PRINT "Enter quantity produced: "
    READ quantity
    PRINT "Enter selling price per item: "
    READ price

    total_cost = material + labour + machine
    revenue = quantity * price
    profit = revenue - total_cost

    PRINT "Total Cost:", total_cost
    PRINT "Revenue:", revenue
    PRINT "Profit:", profit

    IF profit >= 5000 THEN
        PRINT "Profit Target Achieved"
    ELSE
        PRINT "Profit Target Not Achieved"
    ENDIF
END
```

3. Python Source Code, Test Cases, and Outputs

Python Source Code

Python

```
# Business Cost, Revenue, and Profit Analysis System

material = float(input("Enter material cost: "))
labour = float(input("Enter labour cost: "))
machine = float(input("Enter machine cost: "))
quantity = int(input("Enter quantity produced: "))
price = float(input("Enter selling price per item: "))

total_cost = material + labour + machine
revenue = quantity * price
profit = revenue - total_cost

print("Total Cost:", total_cost)
print("Revenue:", revenue)
print("Profit:", profit)

if profit >= 5000:
    print("Profit Target Achieved")
```

else:

print("Profit Target Not Achieved")

Test Cases Table

Test Case ID	Material (Rs.)	Labour (Rs.)	Machine (Rs.)	Quantity	Price (Rs.)	Total Cost (Rs.)	Revenue (Rs.)	Profit (Rs.)	Evaluation Status	Test Status
TC_01	1000.0	500.0	200.0	100	100.0	1700.0	10000.0	8300.0	Profit Target Achieved	Pass
TC_02	2000.0	1000.0	500.0	50	80.0	3500.0	4000.0	500.0	Profit Target Not Achieved	Pass
TC_03	500.0	300.0	200.0	60	100.0	1000.0	6000.0	5000.0	Profit Target Achieved	Pass
TC_04	1500.0	800.0	700.0	40	120.0	3000.0	4800.0	1800.0	Profit Target Not Achieved	Pass

Simulated Terminal Output

Bash

Enter material cost: 1000

Enter labour cost: 500

Enter machine cost: 200

Enter quantity produced: 100

Enter selling price per item: 100

Total Cost: 1700.0

Revenue: 10000.0

Profit: 8300.0

Profit Target Achieved

4. Conclusion

This Python script effectively streamlines business financial evaluation by combining expense summation, revenue multiplication, profit subtraction, and conditional threshold checks. It provides an accurate, automated foundation for monitoring financial performance, making it a valuable tool for small business accounting and enterprise decision-making modules.